Practical 1a: College Information Portal as PWA

AIM: To create a simple website for a College Information Portal and convert it into a Progressive Web Application (PWA) by adding a Service Worker and Web App Manifest.

Steps :

1. Create an HTML/CSS website for the College Portal
2. Create manifest.json with app metadata
3. Create service-worker.js to cache assets
4. Register the service worker in index.html

Index.cshtml

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">

   <meta name="viewport"
       content="width=device-width, initial-scale=1.0">

   <title>College Information Portal</title>

   <!-- PWA Manifest -->
   <link rel="manifest" href="manifest.json">

   <!-- CSS -->
   <link rel="stylesheet" href="style.css">
</head>

<body>

   <header>
      <h1>College Information Portal</h1>
      <p>Welcome to Our College</p>
   </header>

   <nav>
      <a href="#home">Home</a>
```

```
    <a href="#courses">Courses</a>
    <a href="#notice">Notice</a>
    <a href="#contact">Contact</a>
</nav>

<main>

    <section id="home">
        <h2>About College</h2>
        <p>
            Welcome to our College Information Portal.
            This website provides college information,
            courses, notices and contact details.
        </p>
    </section>

    <section id="courses">
        <h2>Courses Offered</h2>

        <ul>
            <li>B.Sc. Computer Science</li>
            <li>B.Sc. Information Technology</li>
            <li>B.Com.</li>
            <li>B.A.</li>
        </ul>
    </section>

    <section id="notice">
        <h2>Latest Notices</h2>

        <p>□ Admission process has started.</p>
        <p>□ Internal examination timetable released.</p>
        <p>□ College annual event will be conducted soon.</p>
    </section>

    <section id="contact">
        <h2>Contact Us</h2>

        <p>Email: college@example.com</p>
        <p>Phone: 9876543210</p>
```

```
        </section>

    </main>

    <footer>
        <p>© 2026 College Information Portal</p>
    </footer>


    <!-- Register Service Worker -->
    <script>
        if ('serviceWorker' in navigator) {
            window.addEventListener('load', function () {
                navigator.serviceWorker.register('/service-worker.js')
                    .then(function (registration) {
                        console.log('Service Worker registered:', registration.scope);
                    })
                    .catch(function (error) {
                        console.log('Service Worker registration failed:', error);
                    });
            });
        }
    </script>

</body>
</html>
```

Style.css

```
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
}

header {
    background-color: #0066cc;
    color: white;
    text-align: center;
    padding: 25px;
}
```

```
nav {
    background-color: #333;
    text-align: center;
    padding: 15px;
}

    nav a {
        color: white;
        text-decoration: none;
        margin: 15px;
    }

main {
    width: 80%;
    margin: 20px auto;
}

section {
    background-color: white;
    padding: 20px;
    margin-bottom: 20px;
    border-radius: 8px;
}

h2 {
    color: #0066cc;
}

footer {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 15px;
}
```

Service-Worker.js

```
const CACHE_NAME = "college-portal-v1";

const ASSETS = [
    "/",
    "/css/site.css",
    "/manifest.json"
];
```

```
self.addEventListener("install", event => {
   console.log("Service Worker: Installing...");

   event.waitUntil(
      caches.open(CACHE_NAME)
         .then(cache => {
            return cache.addAll(ASSETS);
         })
   );
});

self.addEventListener("activate", event => {
   console.log("Service Worker: Activated");
});

self.addEventListener("fetch", event => {
   event.respondWith(
      caches.match(event.request)
         .then(response => {
            return response || fetch(event.request);
         })
   );
});
```

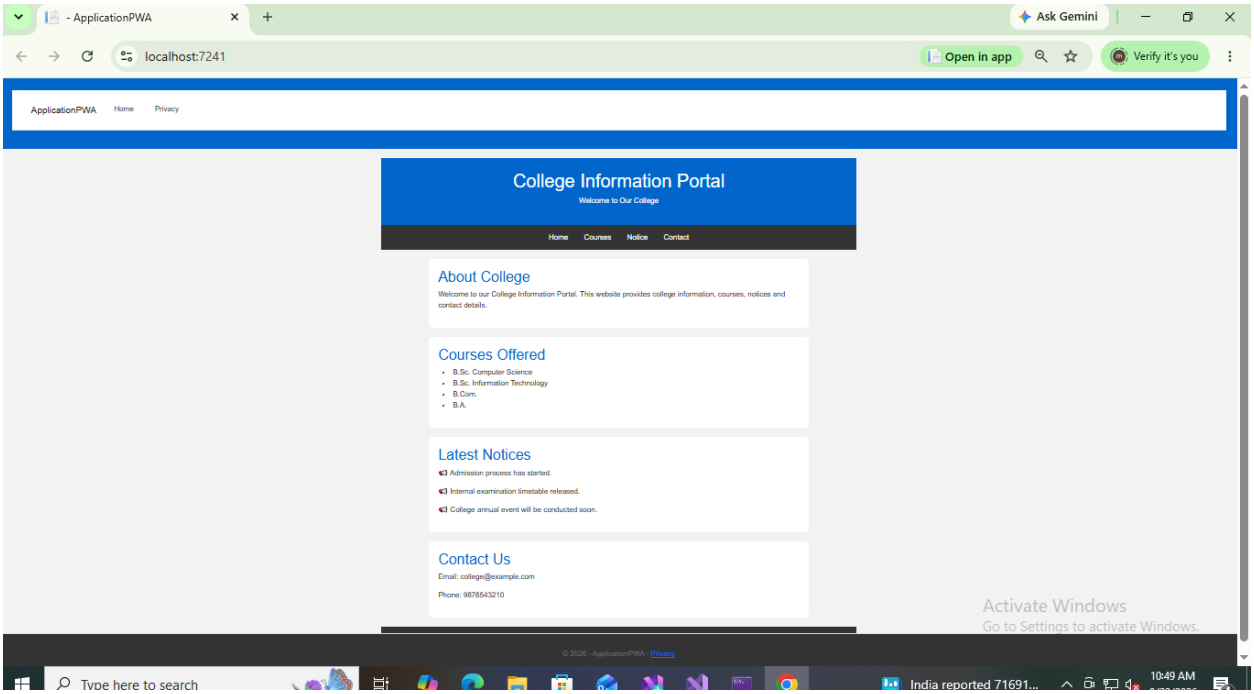
Manifest.json

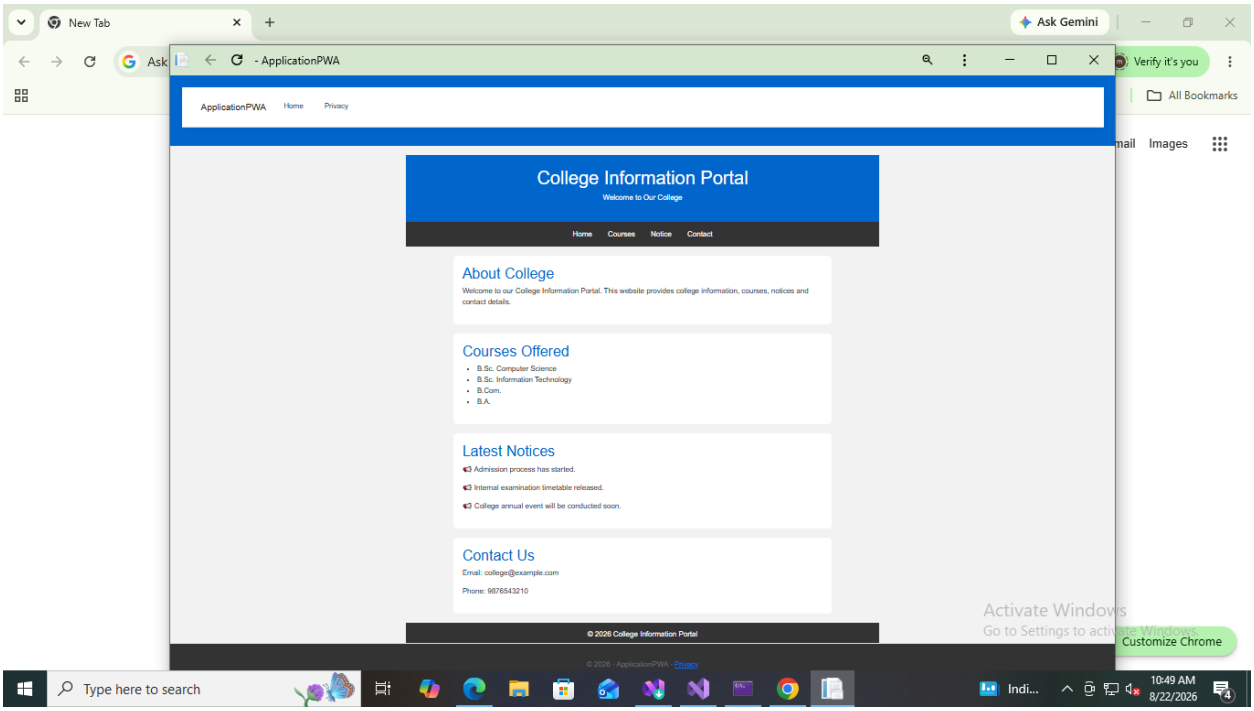
```
{
  "name": "College Information Portal",
  "short_name": "College Portal",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0066cc",
  "description": "College Information Portal Progressive Web Application",

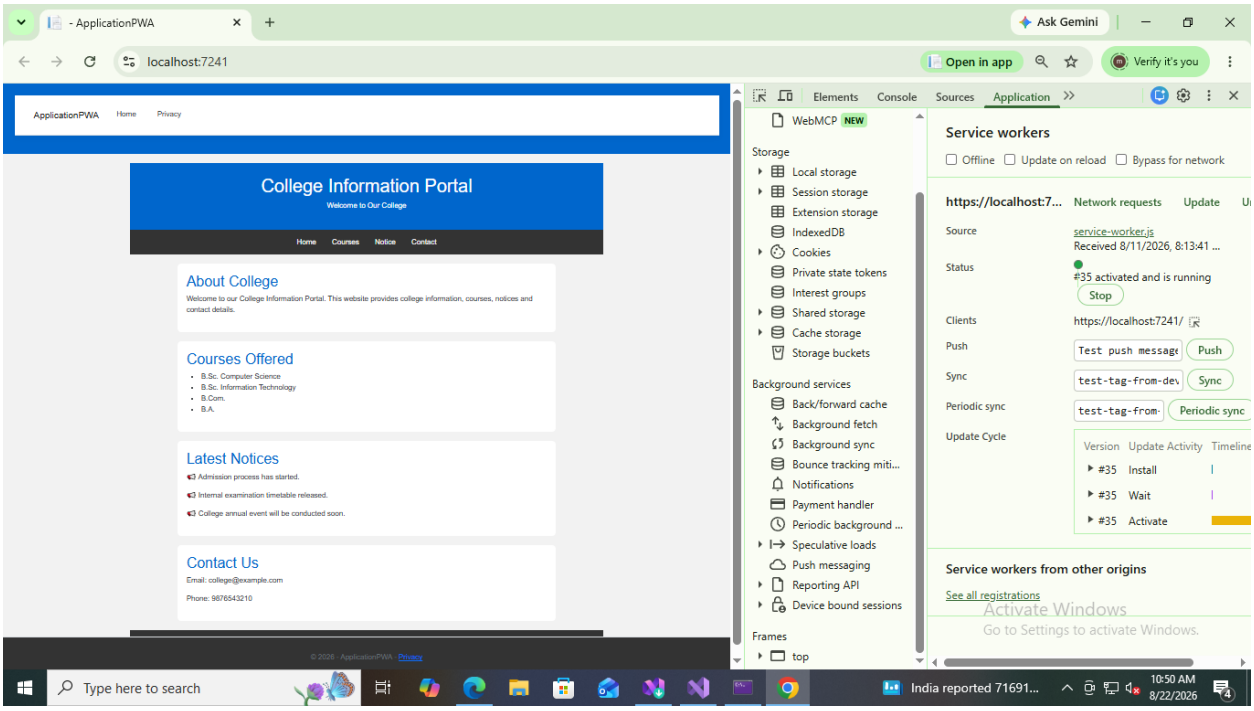
  "icons": [
    {
      "src": "icons/icon-192.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon-512.png",
      "sizes": "512x512",
```

```
    "type": "image/png"
   }
  ]
}
```

Output




New Tab
Ask Gemini
- ApplicationPWA
Verify it's you
All Bookmarks
ApplicationPWA
Home
Privacy
College Information Portal
Welcome to Our College
Home
Courses
Notice
Contact
About College
Welcome to our College Information Portal. This website provides college information, courses, notices and contact details.
Courses Offered
B.Sc. Computer Science
B.Sc. Information Technology
B.Com.
B.A.
Latest Notices
Admission process has started.
Internal examination timetable released.
College annual event will be conducted soon.
Contact Us
Email: college@example.com
Phone: 9876543210
© 2026 College Information Portal
Activate Windows
Customize Chrome
Type here to search
10:49 AM
8/22/2026


- ApplicationPWA
Ask Gemini
localhost:7241
Open in app
Verify it's you
ApplicationPWA
Home
Privacy
College Information Portal
Welcome to Our College
Home
Courses
Notice
Contact
About College
Courses Offered
Latest Notices
Contact Us
Elements
Console
Sources
Application
WebMCP
Storage
Local storage
Session storage
Extension storage
IndexedDB
Cookies
Private state tokens
Interest groups
Shared storage
Cache storage
Storage buckets
Background services
Back/forward cache
Background fetch
Background sync
Bounce tracking miti...
Notifications
Payment handler
Periodic background ...
Speculative loads
Push messaging
Reporting API
Device bound sessions
Frames
top
Service workers
Offline
Update on reload
Bypass for network
https://localhost:7...
Network requests
Update
Source
service-worker.js
Received 8/11/2026, 8:13:41 ...
Status
#35 activated and is running
Stop
Clients
https://localhost:7241/
Push
Test push message
Push
Sync
test-tag-from-dev
Sync
Periodic sync
test-tag-from-
Periodic sync
Update Cycle
Version
Update Activity
Timeline
#35 Install
#35 Wait
#35 Activate
Service workers from other origins
See all registrations
Activate Windows
Go to Settings to activate Windows.
Type here to search
India reported 71691...
10:50 AM
8/22/2026